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Technology opportunity prediction based on SAO representation learning

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Abstract

Current studies on technology opportunity prediction often utilize link prediction on technical element networks, such as co-word networks, patent classification co-occurrence networks, and SAO-based networks. However, these methods typically compute network similarity based on topological structures to predict potential links. They overlook the embedded representations of technical elements and their relationships, which limits the ability to identify semantically related but topologically invisible connections. Moreover, these methods predict combinations of similar technical elements as technology opportunities, but rarely consider combinations of technical elements with multiple relationships as technology opportunities. Therefore, this paper proposes a method to better predict technology opportunities from the perspective of extracting and representing technical Subject-Action-Object (SAO) structures. Firstly, SAO structures are extracted from titles and abstracts of patents using dependency syntax analysis. Secondly, CompGCN model is used to learn entity and relation embeddings by aggregating multi-relational information from the SAO graph. Thirdly, ConvE is designed to capture heterogeneous feature interactions between entity and relation embeddings, and then infers new technical triplets as technology opportunities. The proposed method is validated in the field of speech recognition, the SAOs are extracted and provide a more detailed description of technology compared to keywords. In addition, SAOs that incorporate multiple relations outperform the single relations method with an improvement of 12.3% in MRR, and the method with a graph neural network representation outperforms that without representation learning with an improvement of 2% in MRR. Furthermore, the proposed method ConvE with CompGCN performs best in all prediction models, whose MRR reaches 23.1% and hit@5 reaches 33.6%. Finally, a case study is conducted to further validate the effectiveness of this method by utilizing Subject and Action to predict Object. These results prove that the proposed method is effective and useful. Simultaneously, this method is only a preliminary study and further research is needed in other areas with different multi-relation representation methods.

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Introduction

Predicting potential technology opportunities from a mass of data has been a hot and necessary research topic (Wang et al., 2023). With the rapid and drastic change in technological innovation, capturing technological opportunities becomes a critical factor for countries, businesses, and individuals. Predicting technology opportunities can help researchers in identifying, deriving, and assessing new technological ideas, and provide a scientific decision-making reference for enterprises to optimize innovation paths and formulate innovation strategies (Ma et al., 2021).

The concept of technology opportunity was first proposed by Alan L. Porter (1995), who believed that technology opportunity analysis is to predict the forthcoming technology by mining the horizontal and vertical development trend of existing technologies. The traditional methods of identifying technology opportunities such as Delphi method (Jaemin Cho, 2013), morphological analysis method (Feng et al., 2020; Yoon, 2008; Zhang & Yu, 2020) are mostly qualitative analysis. These methods rely heavily on expert judgment, which can provide valuable insights due to experts' profound domain knowledge and substantial practical experience. However, relying solely on subjective judgment may lead to a certain degree of bias. To overcome this problem, text mining and bibliometric methods were explored. The methods mine technical intelligence from the science, patent data to provide an objective argument for technology opportunity prediction and analysis (Ma et al., 2021). Combining qualitative expert judgment with quantitative analysis can lead to more balanced and robust research outcomes.

Recently, the combination of keyword or patent classification co-occurrence network with link prediction method has been used to predict technology opportunities. This method considers the essence of technology opportunity as a knowledge unit with novelty or a combination of knowledge elements (Ren & Zhao, 2021). However, these keywords or patent classifications just reflect coarse-grained information, and may not fully cover the technical and functional elements of patents. Meanwhile, the correlations between technology elements presented by co-occurrence are not specific enough (X. Li et al., 2023a, 2023b), and cannot reveal how technical elements interact with each other (i.e., based on function or effect relations). Therefore, studies use feature terms or SAO (Subject-Action-Object) to represent fine-grained knowledge units and their various relationships to predict technology opportunities (Choi et al., 2011; Wang et al., 2017; Yang et al., 2018). However, these methods typically assess network similarity in the SAO network based solely on topological structures to predict potential links for technology opportunity prediction. They overlook the semantic information of nodes and the heterogeneous semantic associations among them, which limits the ability to identify semantically related but topologically invisible connections. Embedded SAOs, obtained through representation learning, capture not only the structural information of nodes in the SAO network but also the multiple types of semantic relations among them. Building on these embeddings, calculating the semantic associations between embedded SAOs can improve both the accuracy and the interpretability of technology opportunity prediction. Moreover, the current prediction results of technology opportunity are solely the combinations of similar technical elements (Cao et al., 2023; Han et al., 2021). They rarely predict combinations of technical elements with multiple relationships as technology opportunities.

Therefore, this paper proposes a method based on SAO representation learning for technology opportunity prediction. Firstly, the SAO structures are extracted from patent texts, i.e., S and O represent technical entities that fully capture technical elements in the patents, and A denotes the functional relation between these elements. In this way, a great deal of semantic information about technical elements and their relations can be revealed. Secondly, Composition-Based Multi-Relational Graph Convolutional Networks (CompGCN) model is used to transform entities (S and O) and relations (A) into low-dimensional vector representations whilst maximally preserving properties like graph structure and information. These entity and relation representations are used to predict unseen interactions in SAO structures, to improve the accuracy of subsequent prediction tasks. Finally, the knowledge graph completion ConvE model is used to capture rich heterogeneous feature interactions between technical entity and action relation embeddings to infer new SA-O interactions, and output technical triplets as technology opportunity combinations. So that the obtained technology opportunities not only consist of technical element combinations but also contain technical element relations.

Related works

In this section, we first introduce the related studies on technology opportunity prediction, and then review the research of SAO extraction and mining. At last, the overview of knowledge graph representation learning is presented.

Technology opportunity prediction

Technology opportunity discovery refers to the process of identifying new, underexploited, or high-potential technologies, products or domains that can lead to innovation or competitive advantage (Yoon et al., 2015). The directions of technology opportunity discovery can be broadly categorized into three types (Qiao et al., 2025). The first type focuses on identifying new domains where existing technologies can be applied (Ma et al., 2021; Porter et al., 2020). The second type aims to discover novel technology combinations by analyzing technical elements and their interactions (Han et al., 2021). The third type concentrates on detecting novel or disruptive technologies by examining the temporal dynamics of technological evolution (Liu et al., 2024). In our study, we focus on the second type. This approach aims to predict novel and valuable technical combinations by analyzing fine-grained technical elements along with their multi-relational representation learning. Current studies can be divided into two categories, i.e., patent classification analysis and patent text mining methods.

Patent classification analysis methods often use relationships between patent classifications to predict technology convergence opportunities. Won Sang Lee (2015) predicted technology convergence by applying both association rules and link prediction to the co-occurrence relations of international patent classification (IPC), and further used a topic model to discover emerging convergence technologies. Lee et al. (2021) applied link prediction in the firm-centered F-term network, to recommend technology opportunities for a target firm. Inchae Park (2018) predicted potential technological knowledge flows (TKFs) between heterogeneous fields, and extracted the potential increased knowledge flow links and emerging knowledge flow links as incremental and radical technology opportunities. Yingwen Wu (2023) constructed an enterprise-centered multi-level supply chain IPC

multi-layer network, and recommended IPC for each echelon of the supply chain as technology opportunities. Park and Geum (2022) used three types of features for technology convergence opportunities: GCN features, edge features, and similarity, and finally used portfolio matrix for a firm-level decision-making in technology analysis.

Patent text mining methods often leverage relationships between keywords or technical phases to predict technology opportunities. Liu et al. (2023a) constructed patent-keyword vector matrices for both target and reference technologies, and utilized generative topographic mapping (GTM) and link prediction to predict target technology opportunities. Ren and Zhao (2021) constructed domain knowledge networks (DKNs) based on the syntactic dependencies of technical words, and applied a regression analysis and ant colony optimization to discover the optimal set of technology opportunities. Ma et al. (2019) constructed a network with extracted feature terms and applied link prediction to identify potential research opportunities in the field of gold nanostructures. Han et al. (2021) predicted potential links between technical elements by employing a link prediction algorithm in the SAO network constructed from patent documents. Feng et al. (2022) constructed three stage technology roadmaps (TRM) based on Biterm Topic Model (BTM) and SAO, and used link prediction to identify potential technology opportunities.

In general, most of the related studies use complex network analysis on IPC or keywords for technology opportunity prediction. These methods can provide relatively accurate results at a broader level of granularity. However, because IPC and keywords are not sufficiently fine-grained, they inevitably miss important technical details. Furthermore, these methods simplify the complex relationships between technical elements into co-occurrence relationships, overlooking the heterogeneous, multi-relational semantic associations among them. Therefore, this paper intends to incorporate both technical elements and multiple semantic relationships to explain technology opportunities.

SAO extraction and mining

The SAO structure consists of a subject (noun phrase), an action (verb phrase), and an object (noun phrase). In patent texts, SAO can represent various technical information, where the subject and object can refer to components of a system, and the action can refer to the functionality performed by the components (Choi et al., 2012a, 2012b). Additionally, the SAO can represent technical problem–solution pairs (Garechana et al., 2022).

To extract SAO structures from technical texts, scholars employ dependent syntactic analysis method and natural language processing toolkits such as StanfordCoreNLP (X. Li et al., 2023a, 2023b), Stanford Parser (Liu et al., 2023b; Wang et al., 2022), and GoldFire (Wang et al., 2017). Compared to keyword-based approaches, SAO can reveal semantic relationships between technical elements, presenting more comprehensive technical information. Therefore, it is widely used in current research on technology mining. SAO similarity is used to measure the similarity of patents (Wang et al., 2019), identify patent infringement risks (Kim & Yoon, 2021), and identify technology opportunities with anomaly detection methods (X. Li et al., 2023a, 2023b; Yoon & Kim, 2012). Furthermore, SAO is used to identify "function" or "effect" information (Kim et al., 2019; Yun et al., 2022) and then create technological efficacy matrices (Wang et al., 2022) or technological roadmaps (Choi et al., 2012a, 2012b; Feng et al., 2022). In addition, the SAO networks are used to identify core technological components (Wang et al., 2019), reveal technological evolution trends (Yang et al., 2018), and discover technology opportunities combined with link prediction (Han et al., 2021).

These methods can efficiently and easily extract SAO structure and compute network similarity based on topological structures, such as Common Neighborhoods and the Jaccard Index, to predict potential links. However, they rarely consider the embedded representations of SAOs, which limits their ability to represent the semantic information of nodes and the heterogeneous semantic associations among them. This makes it difficult to identify potential technology combinations that are semantically related but weakly reflected in the network topology. Moreover, these methods identify potential technology opportunities by combining similar technical elements, but they often overlook the combinations of technical elements with multiple relationships. Therefore, this paper explores the use of embedded SAO representations to predict technology opportunities represented by SAO triplets.

Knowledge graph representation learning

The primary objective of knowledge graph representation learning is to obtain embedded representations of entities and relations, which are subsequently utilized in various graph-related tasks such as the technology opportunity prediction task discussed in this paper.

Scholars have developed many scoring functions to train knowledge graph representations, primarily falling into three categories: translation-based models, tensor decomposition-based models, and neural network-based models. In translation-based models, relations are viewed as translations from a subject entity to an object entity. TransE (Bordes et al., 2013) and its variants, such as TransH (Wang et al., 2014), TransR (Lin et al., 2015), and TransD (Ji et al., 2015) are representative translational models. Tensor decomposition-based models use linear or bilinear transformation procedures to compute the semantic similarity scores for entity-relation triplets. The model RESCAL (Nickel et al., 2011), is commonly used to capture the interactions of entities and relations using a bilinear operation. Follow this, a variety of models based on tensor decomposition have been developed, including DistMult (Yang et al., 2014), ComplEx (Trouillon et al., 2016), and TuckER (Balažević et al., 2019). ConvE introduces an innovative approach to modelling entity-relation interactions by utilizing 2D convolutional layers, which enable it to project embeddings into a shared 2D space and capture heterogeneous feature interactions within knowledge graphs (Dettmers et al., 2018). Lin et al. (2024) used ConvE to reshape and convolve embeddings to learn complex entity interactions for knowledge graph completion in railroad maintenance.

Recently, Graph Neural Networks (GNNs) have been introduced to the encoder-decoder framework for learning knowledge graph representations. GCN is an encoder that learns entity and relation representations based on the graph structure, and the scoring function is a decoder that predicts the new links between entities. However, GCNs are not well-suited for multi-relational graphs. To address this limitation, Schlichtkrull et al. (2018) proposed R-GCNs, which incorporate relational information into GNN and design aggregation functions for multi-relations. Vashishth et al. (2019) proposed CompGCN to address the problem of parameter overload in the learning of multi-relational graph representations. Chen et al. (2024) compare CompGCN and SE-GNN, which showed excellent performance on multiple datasets. By combining combinatorial operations with relational modeling, CompGCN can efficiently represent the rich semantics of multi-relational graphs, ensuring more efficient and expressive representation learning.

In conclusion, traditional GCNs can integrate graph structure and node features for representation learning, but they are not well-suited for multi-relational graphs due to their

inability to model different types of relations. In contrast, the multi-relational graph neural network model CompGCN can effectively represent the multi-semantic relationships and heterogeneous interactions among complex technical elements during the embedding of entities and relations. Therefore, this paper adopts the multi-relational graph neural network model, CompGCN, to learn SAO representations and utilizes the scoring function, ConvE, for predicting technology opportunities.

Data and method

The method for predicting technology opportunity includes four parts, as shown in Fig. 1. Firstly, patent documents are collected and preprocessed to obtain patent titles, abstracts, and patent numbers in a specific field. Secondly, SAOs are extracted from the titles and abstracts of patents using dependency syntax analysis method. Thirdly, the CompGCN model is used for SAO representation learning, leveraging composition operations to jointly embed both nodes and relations in SAO graph. Finally, the ConvE model concatenates entity and action embeddings and processes these features into a fully connected layer, to predict technical triplets and their scores as technology opportunities.

Data description

With the development of artificial intelligence, the use of speech recognition is expanding rapidly, driven by continuous technological advancements and strong market demand. This field is highly integrated with other domains, reflecting the potential for technological convergence and innovation (Bhatt et al., 2021). Therefore, this study selects the field of speech recognition as the data source and collects data from the Derwent Innovations Index Patent Database (DII).

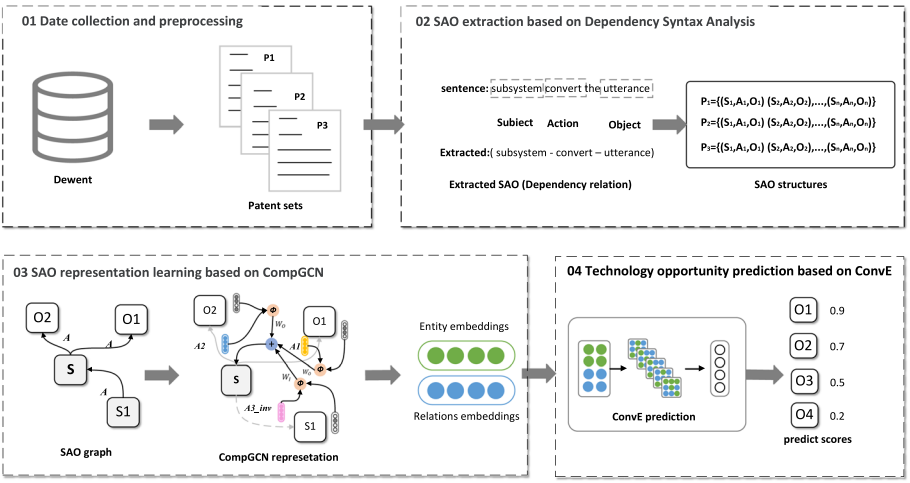


Fig. 1 Framework of the method

Table 1 Patent retrieval formula for speech recognition

Field	Retrieve query
Subject	TI = (speech-recogin*or"speech recogni*" or (speech NEAR/1 recogni*) OR ASR) OR AB = (speech-recogin*or"speech recogni*" or (speech NEAR/1 recogni*) OR ASR)
MAN	MAN = (T01-J18 OR T01-C08A OR W01-C01 Q4 OR W01-C01 B5 B OR W01-C01Q8C OR W02-G02A5C OR W02-K06 OR W03-A02C5L OR W04-V01 OR W04-V04 OR W04-V05 OR W04-V09 OR X22-L OR W03-A18A6 OR W03-B08C3 OR W03-B08 C8)
IPC	IP = (G10L-015/00 OR G10L-015/01 OR G10L-015/02 OR G10L-015/04 OR G10L-015/06 OR G10L-015/08 OR G10L-015/20 OR G10L-015/22 OR G10L-015/24 OR G10L-015/26 OR G10L-015/28)

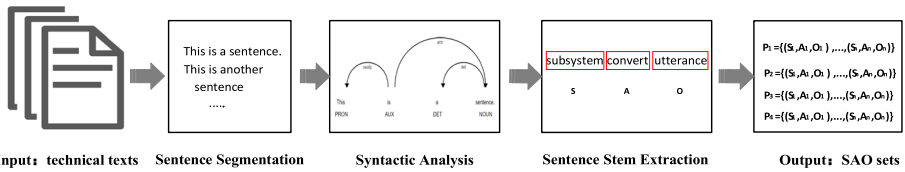


Fig. 2 The process of extracting SAO structures

We refer the previous query construction strategy and build the retrieval query by combining domain-specific keywords and specific classification codes reflected in the IPC and MAN fields (Clarke et al., 2020). The specific patent retrieval query for the technological field of speech recognition is shown in Table 1, and the retrieval time range is from 2017 to 2021. After excluding irrelevant, mismatched, duplicate, and missing data, a total of 6490 patents are obtained, including the titles, abstracts, patent numbers, and years of application.

SAO extraction based on dependency syntax analysis

SAOs are commonly used to represent functions of technology. They consist of "Subject" (noun phrase), "Action" (verb phrase) and "Object" (noun phrase), and explicitly describe a relationship between components that appear in a technical text. The dependency syntactic analysis method can effectively identify these "subject-verb-object" structures in sentences. Therefore, this study employs the dependency syntactic analysis method to extract SAO structures from the text, as shown in Fig. 2.

The process of SAO extraction is as follows. Firstly, technical texts such as patents are chosen as the corpus, which are preprocessed and segmented into individual sentences based on punctuation. Secondly, the sentences are lexically annotated using a syntactic parser (spaCy[1]), and their syntactic constituents are analyzed in terms of their dependency. Thirdly, according to the subject-predicate-object setting extraction rules, S and A are extracted based on labels like "nsubj" (noun subject), "csubj" (subordinate subject), or "nsubjpass" (passive subject) in the sentences. Similarly, A and O are extracted based on

[1] https://gitee.com/mirrors/spacy

labels “dobj” (direct object) or “iobj” (indirect object). These phrases are then combined to form a complete SAO structure. An example of SAO structure extraction is illustrated in Fig. 3.

In this example, we can extract SAO triplet “(automatic speech recognition subsystem, convert, utterance)” from the sentence. In addition, technical entities usually involve types of components, systems, algorithms, models, application domains, and effects, and the corresponding relationships include aspects of composition, effect, and function.

However, the initial SAO may exhibit problems such as tense inconsistency, missing or duplicated structures, and complex sentences within technical entities. To ensure complete semantics of SAO, it is necessary to perform deduplication, removal of stop words, lemmatization, and simplification of technical entities using keybert (Nils Reimers, 2019).

SAO representation learning based on CompGCN

This section includes both SAO graph construction and entity-relation representation. Firstly, the SAO triples are transformed into a graphical form. Then, the CompGCN model is utilized to embed the entities and relations into semantic vectors. These embeddings serve as inputs for the subsequent technology opportunity prediction task.

SAO graph construction

SAO triples need to be converted into a knowledge graph format. We define the SAO graph as $G = (E, A, D, H)$, where E and A represent the collections of entities and relations, respectively, while D and H indicate the initial feature vectors for the corresponding entities and relations. In the SAO graph, S and O are depicted as nodes, whereas A is represented as edges, as illustrated in Fig. 4.

It can be observed that the SAO triples are ultimately transformed into nodes and edges. The nodes denote the modules, units, algorithms, and effects in a technology, and the edges denotes the relations between two nodes, e.g., “use” relation between nodes “module” and “algorithm”, the “improve” relation between node “module” and “effect”, and the “comprise” relation between node “unit” and “module”.

Entity-relation representation

CompGCN can aggregate node and edge information through composition operations, and jointly learn the embeddings of entities and relations. This makes it particularly suitable for tasks involving complex relational structures. Therefore, we adopt CompGCN to learn embeddings for SAO. The framework of SAO representation learning based on CompGCN

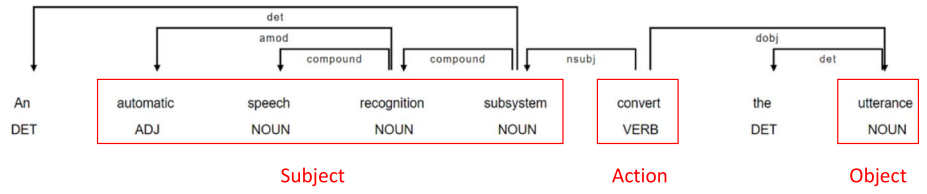


Fig. 3 The example of SAO structure extraction

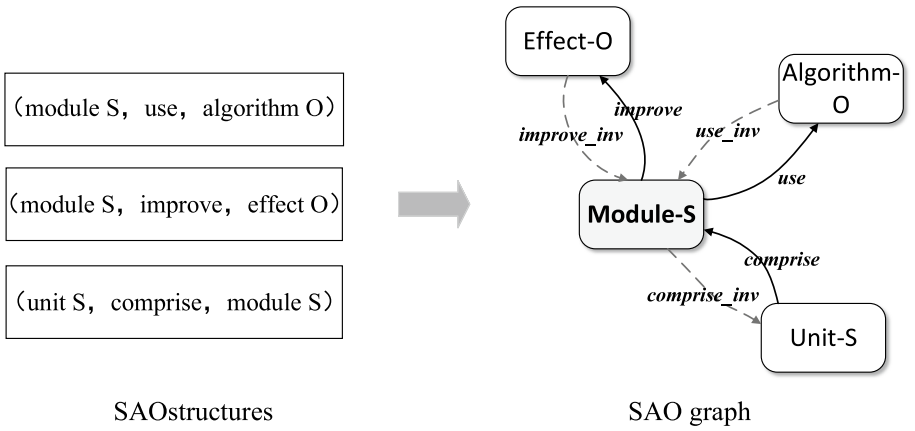


Fig. 4 Converting SAO structure to a SAO graph

is shown in Fig. 5, where SAO graph is used as input. For actions representation, they are transformed using a separate weight matrix. For S and O representation, a composition operation (ϕ) is applied to each edge in the neighborhoods of the central node (e.g., Module-S in Fig. 5). These composed embeddings are then convolved with specific filters (W_O), (W_I), and (W_S) to construct original relation, inverse relation, and self-relation respectively. This message from all the neighbors is then aggregated to get the updated embedding of the central node. The final output consists of the embeddings for the entities and action relations. In addition, we omit self-relation in the Fig. 5 for clarity.

Relation representation To prevent excessive parameterization, CompGCN transforms the initial feature vectors of relations into a set of vectors in the initial convolutional layer. The relation vector h_a represents by Eq. 1.

$$h_a = W_{action} Z_a \tag{1}$$

In Eq. 1, $W_{action} \in \mathbb{R}^{d_1 \times d_0}$ is a learnable transformation matrix that projects all the relations to the same embedding space as nodes, and z_a denotes initial features for action relation a. Typically, the initial values of z_a can be randomly initialized numerical vectors

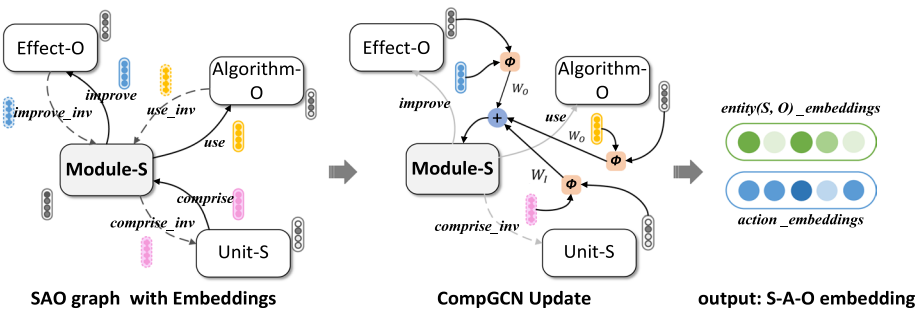


Fig. 5 SAO representation learning based on CompGCN

ranging from − 1 to 1, which are usually sampled from a uniform or normal distribution. Subsequently, the vector is then utilized as the input of the next CompGCN layer.

Entity representation In each convolutional layer, entities in CompGCN update their embeddings by aggregating heterogeneous information from their neighboring entities. For example, to update the embedding of the "module" node, it is necessary to aggregate the semantic information from its adjacent entities (e.g., "algorithm", "device", and "effect"), original adjacency relations (e.g., "module—*use*—algorithm", and "module—*improve*-effect"), reverse adjacency relations (e.g., "module—*comprise_inv*—unit"), and self-related relation (e.g., "module—module"). This aggregation generates an embedding vector for "module S" that contains rich semantic information. The aggregated feature vectors h_v are generated by Eq. 2 as follows.

$$h_v = \sum_{(u,a)\epsilon\mathcal{N}(v)} W_{\lambda(a)}\phi\left(x_u, z_a\right) \tag{2}$$

In Eq. 2, x_u and z_a denotes initial features for node u and action relation a respectively, and h_v denotes the updated representation of node v. Here, the combination operation ϕ for relations and entities is performed using multiplication (Mult) or circular convolution (Corr), as shown in Eq. 3 and 4.

$$Multiplication(Mult) : e_o = \phi\left(e_s, e_a\right) = e_s \times e_a \tag{3}$$

$$Circular - correlation(Corr) : e_o = \phi\left(e_s, e_a\right) = e_s \times e_a \tag{4}$$

Here, $\phi : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d$ is a composition operator, where s, a and o represent the subject, action, and object in the knowledge graph, and $e(\cdot) \in \mathbb{R}^d$ represents their corresponding embeddings. In the subsequent experiments, we will train the model using these two methods separately and select the one with the best performance for prediction.

Furthermore, $W_{\lambda(a)}$ represents the specific parameter weight matrices for edge relation types, and λ is the specific weight of several directions, as shown in Eq. 5.

$$W_{\lambda(a)} = f(x) = \begin{cases} W_O, a \in A \\ W_I, a \in A_{inv} \\ W_S, a = T(self - relation) \end{cases} \tag{5}$$

In Eq. 5, O, I, and S respectively represent the original direction, reverse direction, and self-relation direction.

SAO updating representation We can extend the formulation of Eq. 2 to the case where we have k-stacked CompGCN layers. Let h_v^{k+1} denote the representation of a node v obtained after k layers which is defined in Eq. 6.

$$h_v^{k+1} = f\left[\sum_{(u,a)\epsilon\mathcal{N}(v)} W_{\lambda(a)}^k \phi\left(h_u^k, h_a^k\right)\right] \tag{6}$$

Similarly, let h_v^{k+1} denote the representation of action a after k layers, as shown in Eq. 7.

$$h_a^{k+1} = W_{action} h_a^k \tag{7}$$

Here, h_v^0 and h_a^0 are the initial node x_u and action z_a features when $k = 0$, respectively.

The final output of the representation vectors of entities S, O, and A is used as input data for the subsequent prediction task.

Technology opportunity prediction base on ConvE

In this study, we believe that previously unseen but strongly correlated entity-relation triplets, which are predicted based on the existing SAO graph, have the potential to become future technology opportunities. ConvE has demonstrated excellent performance in knowledge graph completion tasks, reflecting its effectiveness in predicting unseen relations. Its ability to infer latent connections directly is well-suited to our goal of discovering innovative technology opportunities from the SAO graph. Besides, the inherent capability of ConvE to handle multi-relational data makes it naturally compatible with our framework. Therefore, we employ the ConvE model to predict unseen SAO structures. This model utilizes a two-dimensional convolution to capture heterogeneous feature interactions between entity and relation embeddings. Through a scoring function, it can infer new technical triplets as technology opportunities. For example, v is a component entity (S) and a is the "device—*contain*—component" relation (A), and the ConvE model takes node embeddings Z_v and relation embeddings Z_a as input to predict another entity u (O). This process predicts another entity with high score given a specific entity and relation, as shown in Fig. 6.

For a technical triple (e_s, a, e_o), the scoring function is defined in Eq. 8. In the equation, the entity embeddings $\overline{e_s}$ and the relation embeddings $\overline{a}$ are learned by CompGCN representation learning module, which are concatenated as input to a 2D convolutional layer. This layer returns a feature map tensor $\tau \in \mathbb{R}^{c \times m \times n}$, where c represents the number of 2D feature maps with dimensions m and n.

$$\hat{y} = \varnothing(e_s, a, e_o) = f\left(vec\left(f\left(\left[\overline{e_s}; \overline{a}\right] * w\right)\right)W\right)e_o \# \tag{8}$$

Finally, the feature map tensor τ is compressed into a one-dimensional vector $vec(\tau)$. This vector then undergoes a linear transformation parametrized using the matrix $WW(0,1)$ probability by applying the sigmoid function. Based on the probabilistic scoring results, the model can predict potential technology opportunities by identifying the target entity e_o in the missing triplet.

For training the model parameters, we use a binary cross-entropy loss function based on negative sampling, so that the trained model has a higher score for positive samples than negative samples. For each positive sample triplet (e_s, a, e_o), the model randomly replaces the head entity or tail entity of the triplet to generate a negative sample triplet (e_s, a, e'_0) or

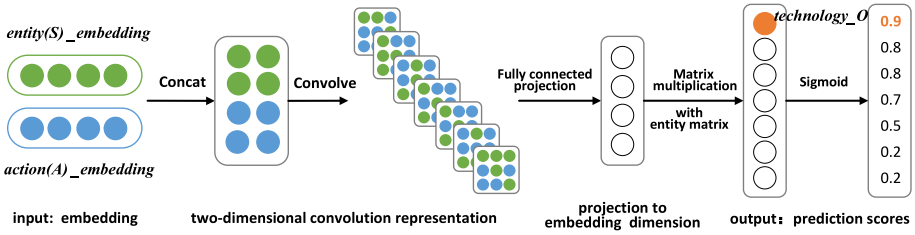


Fig. 6 ConvE model used for technology opportunity prediction

(e'_s, a, e_0), where the entity e'_0 is unrelated to e_s, a, and e_o, and e'_s is unrelated to a, e_o. The binary cross-entropy loss function is calculated as shown in Eq. 9:

$$Loss = -\frac{1}{N}\sum y log\hat{y} + (1-y)\cdot \log\left(1-\hat{y}\right) \tag{9}$$

In Eq. 9, y represents the true label of the sample, with 1 indicating positive sample and 0 indicating negative sample, and $\hat{y}$ represents the triplet scores outputted by the encoder.

Evaluation base on knowledge graph completion

In our study, we treat the evaluation as a knowledge graph completion (KGC) process, which evaluates the quality of a predictive model by comparing the ranked results of the predicted technology triplet scores with the correctly ranked results. Following previous KGC experiments (Gao et al., 2022; Z. Li et al., 2023a, 2023b), several widely used evaluation metrics including MRR and Hits@n are adopted to measure performances of technology opportunity prediction experiments.

MRR, short for Mean Reciprocal Ranking, calculates the average reciprocal rank of all correct triplets in the ranking results. The larger the value of this metric, the better the model performance. The calculation method is shown in Eq. 10.

$$MRR = \frac{1}{|S|}\sum_{i=1}^{|S|}\frac{1}{rank_i} = \frac{1}{|S|}\left(\frac{1}{rank_1} + \frac{1}{rank_2} + \cdots + \frac{1}{rank_{|S|}}\right) \tag{10}$$

Hits@n, also known as the hit rate, is a metric used to evaluate model performance, indicating whether the target outcome appears within the first n predictions. In our study, it calculates the proportion of correct triplets in the top n ranking results. The larger the value of this metric, the better the model performance. *Hits@n* is calculated as shown in Eq. 11. If $rank_i \le n$, the value is 1, indicating that the correct triplet i is ranked within the top n positions; otherwise, the value is 0. Referring to previous studies, we choose n as 1, 3, 5, 10 and 20 to evaluate the performance of the model with different top n values (Yu et al., 2023; Zhang et al., 2025).

$$Hits@n = \frac{1}{|S|}\sum_{i=1}^{|S|}\left(rank_i \le n\right) \tag{11}$$

Experiments and results

The experiment consists of four parts to validate the effectiveness of the method. First, the result of extracted SAOs is compared with keywords which also extracted from the same patent texts. Second, the CompGCN model is used to transform technical entities (S and O) and relations (A) into low-dimensional vector representations, respectively. Third, we conduct three comparisons for technology opportunity prediction. The first compares the results between SAO with multiple actions and SO that treat all different actions in SAO into one same action. The second compares the results with or without multi-relation representation learning. The third compares the performance of different prediction models. Finally, we analyze the predicted technology opportunities through a special case study.

Experimental setup

The relevant settings of experiment are shortly summarized before introducing the results. Firstly, we construct a dataset and divide it into training and testing set by different time periods. Then, we set the parameters of the models involved in the experiment. Finally, we introduce the evaluation metrics for the experiment.

Experimental datasets construction

SAO structures from 2017 to 2021 are selected as the experimental dataset. The data from 2017 to 2020 is used as training dataset for model training and the data in 2021 is used as the testing dataset for prediction evaluation. Notably, there are duplicate data between the testing dataset and the training dataset. Therefore, we remove these duplicated data from the testing dataset. This allows us to validate the previously unassociated combinations in the training set, where predictions may be correlated. In the initial dataset, 26,714 SAO triples are obtained, and after removing duplicates, 23,479 unique SAO structures are obtained. Then, technical entities with low frequency of occurrence are excluded, and technical entities and relations with more than 3 occurrences are retained, to construct experimental datasets. Finally, a total of 2,994 entities (S and O), 239 actions, and 7,653 SAO triplets are obtained. Table 2 shows the details of the experimental dataset.

Parameter setting

We explain the parameter settings of the model in detail from two aspects respectively, i.e., SAO representation learning experiment and technology opportunity prediction experiment.

In terms of experiment on SAO representation learning, we establish the model CompGCN with batch_size of 128 (Vashishth et al., 2020). The GCN layer is set to 1, as we find that deeper GCN layers do not improve the performance, but rather increase the computational cost significantly. The embeddings dimension is set to 200, which is a trade-off between the training performance and computational cost. Moreover, a learning rate of 0.001 is set to facilitate effective model training. To avoid overfitting, the dropout setting after each convolution layer is set to 0.1 in the embedding module.

As for technology opportunity prediction, the ConvE (Dettmers et al., 2018) model is used on experimental dataset. We set the embedding size as 200 to be consistent with the CompGCN embedding layer. The dropout with a value of 0.1 and 0.3 are applied to the feature maps and hidden features in structural features learning. During training, the batch size is 128, the learning rate is 0.001, and label smoothing is set as 0.1.

Table 2 Description of experiment dataset

	Period	Number of patents	Number of SAO
Training dataset	2017–2020	5634	7266
Testing dataset	2021	775	387

Table 3 The top 10 most frequent extracted SAO structures

ID	S	A	O	Count
1	Multiple techniques	Identify	Error condition	23
2	Speech recognition engine	Receive	Waveform output	19
3	Model training	Set	Sequence states	19
4	Voice recognize device	Obtain	Characteristic matrix data	17
5	Identification unit	Send	Voice recognition module	14
6	Simple structure	Generate	Voice recognition	14
7	Signal processing apparatus	Set	Acquisition unit	13
8	Optimal acoustic model	Improve	Accuracy recognition results	12
9	Input decoder	Base	State sequence	10
10	Accuracy of detection	Set	Threshold value	8

SAO structures extraction and analysis

In this section, we perform SAO structure extraction and select the top 10 occurrences of the number of SAO for analysis. Then, we compare the results of the SAOs and keywords extracted from the same patents.

SAO structures extraction results

We utilize the natural language processing toolkit Spacy to extract SAO structures from the titles and abstracts of patents. Then, SAO structures are preprocessed, including removing missing data, stop words, lemmatization, and semantic deduplication. In addition, KeyBert (Nils Reimers, 2019) method and TextBlob[2] are further used to filter and obtain technical entities containing complete semantic information. Finally, 26,714 SAO structures are obtained, and after removing duplicates, 23,479 unique SAO structures with different expressions are obtained.

The top 10 most frequent SAO structures are shown in Table 3. The first column ID is the serial number of SAO structures, and the column "S", "A", and "O" represent "Subject", "Action", and "Object", respectively. "Count" indicates the number of SAO structures.

From the table, most of SAO structures denote different components and operations, which are used in speech recognition and speech control systems. For example, the first SAO "multiple techniques identify error condition" indicates that multiple techniques are used in the speech recognition system to detect and recognize error conditions. It may include error detection algorithms, anomaly detection algorithms, etc. The second SAO "speech recognition engine receives waveform output" indicates that the speech engine is the component used to convert speech into text. It receives waveform output from other components and then converts it into recognizable text. The third SAO "model training set sequence states" describes the operation of setting sequence states during the model training. These SAO structures allow us to better understand the technologies and system components used in the field of speech recognition, as well as the relationships between them.

[2] https://github.com/sloria/TextBlob

Comparison of SAOs and keywords in the same patents

To illustrate the superiority of SAO structures compared to keywords in representing technology, this paper performs SAO extraction and keyword extraction on the same patent, where keyword extraction uses Textblob. We randomly select three patents "US2022005457-A1", "US2021225389-A1" and "CN112764535-A" as examples. The results are shown in Table 4, in which "ID" denotes the patent number, "SAO" denotes the SAOs extracted from patent abstract using dependency syntax, and "keywords" denotes the key technical phrases extracted from the patent abstract using the Textblob method.

We can see that the main content of the first patent is the neural network training method for text-to-speech (TTS) and speech-to-text (ASR). Through the keywords, we know that it involves TTS technology, neural networks, audio samples, text samples, and ASR. From these keywords, we can roughly understand that the theme of this patent is to train audio or text data for speech recognition by using methods such as neural networks. While through the SAO structures, we can know that the TTS neural network is used to adjust parameters to optimize the quality of synthesized speech. It uses audio samples for training, converting text to corresponding audio output. The ASR neural network receives audio data and outputs corresponding text to perform speech recognition. Through these, we can not only know that the theme of the patent is speech synthesis technology, but also understand the relationships between each technical phrase more specifically. The SAO form can explain the roles between the technologies.

In general, with the use of keywords, we can only understand the technology applied in a patent, but not understand the relationships between these technologies. However, the SAO structures with multiple actions allow us to understand the interactions between technical knowledge, and to describe technology opportunities in a more detailed and clear manner. Thus, SAO can reveal how technical elements interact with each other and further present the specific relationships of predicted technical element combinations.

SAO representation learning

The extracted SAO are constructed as SAO graph, and entities and relations as initial features are inputted into the CompGCN model. The model is trained to output semantic vector representations of technical entities (S and O) and relations 1(A), as shown in Fig. 7. The first column denotes the extracted entities or actions, while the other columns together represent the semantic embeddings of the words in the first column.

The results show that CompGCN could convert both technical entities and action relations into 200 dimensional vectors ranging from − 1 to 1. Each number in these vectors represents a feature in the semantic dimensions of the SAO, incorporating information about nodes, neighboring nodes, and connecting actions. Moreover, the closer the numbers are in each dimension, the stronger the association between words. Therefore, vector similarity can be used to indicate correlations between entities or relationships and to support subsequent technology opportunity prediction tasks. As shown in Table 5, the top 10 entity pairs and relation pairs are listed, ranked by semantic similarity.

For example, the entity "recognition model" is represented by the vector "[− 0.15490, − 0.23140, − 0.96250, …]", and the entity "acoustic model" is represented by the vector "[0.00820, − 0.30670, − 0.96340, …]". As shown in Table 5, the semantic similarity between these two entities is 0.76844. The high correlation

Table 4 Comparison of SAOs and keywords in the same patents

ID	SAO	Keywords
1	(TTS neural network, adjust, parameters), (the TTS neural network, use, third audio sample)	TTS, neural network, audio sample
	(A first audio sample, reproduce, the first text sample), (second text sample, reproduce, the first audio sample)	Audio sample, text sample
	(Second text sample, use, ASR neural network), (ASR neural network, receive, second audio sample)	ASR, audio sample, text sample,
2	(confidence score, indicate, the probability of correct transcription of the utterance)	Confidence score, correct transcription, utterance
	(The pronunciation score, indicate, extent)	Pronunciation score
	(A confidence score, base, the generating of the intelligibility score), (intelligibility score, represent, an intelligibility of the utterance)	Intelligibility score, confidence score
	(Acoustic signal, encode, utterance of a user)	Acoustic signal, utterance
	(ASR module, generate, multiple best output)	ASR, multiple-best output
	(Multiple-best output, comprise, multiple recognition results)	Multiple-best output, multiple recognition results
3	(Algorithm processing module, connect, output module)	Algorithm processing module, output module
	(Algorithm processing module, optimize, translation result)	Translation result
	(The micro processing module, connect, algorithm processing module), (translates information, connect, micro-processing module)	Micro processing module, algorithm processing module
	(Information collection module, obtains, data input)	Information collection module, data input

semantic representation of entities										
asr_module	-0.32990	-0.34960	-0.83740	-0.25130	-0.68910	-0.41010	-0.15910	-0.43920	-0.86860	-0.75960
wide_variety_devices	-0.26180	-0.33906	0.00028	-0.81631	-0.83017	-0.80418	-0.73818	-0.74521	-0.78350	-0.62113
hand_held_devices_systems	-0.19590	-0.26590	-0.88920	-0.81610	-0.60150	-0.56990	-0.76160	-0.45690	-0.69840	0.05590
speech_recognition	-0.44880	-0.18150	0.71710	-0.83420	-0.11670	-0.93730	0.03620	-0.73700	-0.61100	0.77720
computers_devices	-0.10280	-0.31790	-0.90800	-0.80650	-0.52250	-0.49240	-0.57790	-0.84500	-0.62830	-0.44730
network_server	-0.33070	-0.24960	-0.77750	0.10250	-0.54990	0.07750	-0.34970	0.64740	-0.33620	-0.52250
voice_instruction	-0.07770	-0.34930	-0.80450	-0.36070	0.10570	-0.63180	-0.23240	-0.71470	-0.51730	-0.94760
recognition_model	-0.15490	-0.23140	-0.96250	-0.92240	-0.34280	-0.89750	-0.54620	-0.95620	-0.65450	0.68800
speech_input	-0.26380	-0.22100	0.17840	-0.06090	-0.55420	-0.11230	-0.24040	0.61860	-0.60590	0.43310
speech_data	0.15040	-0.27640	-0.99530	0.36540	0.09770	-0.90840	-0.07930	-0.59470	-0.62000	0.42200
semantic representation of relations										
train	0.08860	-0.03870	-0.21620	-0.29620	0.11300	-0.25120	0.29400	-0.17720	-0.05890	0.02350
compare	0.08591	-0.02835	-0.24998	-0.34430	0.02319	-0.54995	0.20449	-0.10369	-0.10687	0.17288
identify	-0.15420	0.00890	0.38670	0.33220	-0.06030	0.00270	0.27200	-0.20260	0.27510	-0.12910
detect	0.10220	-0.17350	-0.10240	-0.02810	-0.01770	-0.03620	0.31200	-0.22520	-0.05900	0.18020
store	0.15075	-0.04304	0.21446	-0.22056	0.03367	0.09233	-0.03725	-0.53260	0.18115	-0.13137
obtain	-0.18861	0.26875	0.63480	0.31072	0.19800	0.03795	0.06096	0.05680	0.29673	-0.03281
use	-0.32170	0.07780	0.30950	0.32790	-0.09420	-0.30170	-0.01410	-0.28390	0.28300	0.00100
adjust	-0.17560	-0.17203	-0.09364	-0.09067	0.08415	-0.27370	0.14762	-0.21052	0.24530	-0.23937
output	0.10090	0.07600	-0.25880	-0.07760	-0.17130	-0.17670	-0.08490	-0.09310	-0.01170	-0.09530
utilize	0.09000	-0.24080	0.00590	-0.03560	-0.14840	0.09900	0.19690	-0.02650	0.32710	-0.08790

Fig. 7 Examples of semantic representation of entities and relations

Table 5 Top 10 entity/relation pairs ranked by semantic similarity

Ranking	Entity/relation	Entity/relation	Semantic similarity
1	Recognition model	Acoustic model	0.7684408137167659
2	Target audio	Recognition terminal	0.7677647173321108
3	Process	Analyze	0.7574382869106155
4	Context vector	Processing operation	0.7520064404610938
5	Audio signal	Speech segment	0.7489606297665593
6	Obtain	Perform	0.7406282257490283
7	Speech signal	Audio data	0.7365036486977862
8	Control circuit	Control device	0.731631467752273
9	Audio signal	Extraction unit	0.7272143614848527
10	Speech data	Initial recognition model	0.720950924684809

between their vectors indicates a strong semantic association between the entities. Based on this, the original SAO "recognition model process speech signal" can generate a new SAO "acoustic model process speech signal" by replacing the subject (S). This substitution may represent a potential technical opportunity as it explores a related entity within the same action-object context.

Similarly, the semantic similarity between the actions "process" and "analyze" is 0.75744. The high correlation between them allows for the replacement of the action (A) in the SAO "recognition model process speech signal" to generate "recognition model analyze speech signal". Furthermore, the high correlation of 0.73650 between the objects "speech signal" and "audio data" enables the replacement of the object (O) in the SAO "recognition model process speech signal" to form "recognition model process audio data". This substitution can extend technical application scenarios by modifying the processed object. By replacing any one or more elements (S, A, or O), new SAO combinations can be generated. These combinations may indicate potential technical opportunities, offering novel insights for technological innovation.

Comparison of technology opportunity prediction results

We conduct three comparative experiments for technology opportunity prediction. Firstly, we merge multiple actions into a single action, which we refer to as SO for convenience, and compare the effects of multiple actions versus a single action on technology prediction. Secondly, we compare the results predicted by ConvE with or without multi-relational representation, to explore the effectiveness of the multi-relational representation model. Thirdly, we compare the results predicted by different knowledge graph completion models to explore the impact of different predictive models on the results.

Comparison of SAO and SO structures for technology opportunity prediction

To verify the effectiveness of adding multiple actions compared to single-dimensional action, this study conducts a comparative experiment on multiple actions (SAOs) and one type of action (SOs). Accordingly, we obtain two experimental datasets: (1) SAO experimental dataset, which contains 239 actions; (2) SO experimental dataset, which contains a single action formed by merging multiple actions into one. The proposed method CompGCN + ConvE in our study is applied to these two datasets. The result is shown in Table 6, in which the highest value in each column is highlighted in italics.

The result shows that the performance of model adding multiple actions is better than the single action. Specifically, the multiple actions prediction outperforms the single action prediction by12.3%, 10.1%, 15.9%, and 19.6% in terms of MRR, hits@1, hits@3, and hits@5, respectively. It means that these actions can offer additional contextual information for entity representation and assist the model in understanding the relationship between entities at a higher level. While replacing all relations with a single type of relation would result in the loss of the richness of the original data, limiting the learning ability of the model. Therefore, compared to the previous approach using single action, this study could enhance the accuracy of technology opportunity prediction by incorporating multiple actions.

Comparison on models with or without multi-relation representation learning

To validate the effectiveness of using multi-relational graph neural networks CompGCN for learning embedding vectors of entities and multi-relations, this study first compares the results of CompGCN + ConvE and ConvE, where the latter does not incorporate multi-relational representation learning. In addition, we compare CompGCN + ConvE and RelGCN + ConvE to further validate the advantages of CompGCN in multi-relational representation learning. Both RelGCN and CompGCN belong to multi-relational graph neural networks and can embed entities and multiple relations. In the RelGCN + ConvE model and CompGCN + ConvE model, the SAO representation is first learned using multi-relational graph neural networks, and then these learned entity and relation embedding vectors are inputted into ConvE model for prediction. In contrast, the ConvE model removes multi-relational graph neural networks

Table 6 Performance comparison on multiple actions and single action

Type of relations	MRR	hits@1	hits@3	hits@5	hits@10	hits@20
Single action	0.108	0.056	0.112	0.140	0.219	0.322
Multiple actions	*0.231*	*0.157*	*0.271*	*0.336*	*0.406*	*0.493*

and makes direct predictions. The result is shown in Table 7, in which the numbers emphasized in italics correspond to the highest value in each column.

It shows that models combining multi-relational graph neural networks outperform the ConvE alone in terms of MRR and hits@n. Specifically, the RelGCN + ConvE model improves MRR by 0.7%, hits@1 by 0.7%, and hits@3 by 1.8%, while the CompGCN + ConvE model achieves a 2% improvement in MRR, a 2.8% improvement in hits@1, and a 4.6% improvement in hits@3. The reason maybe that these models are composed of a graph neural network as an encoder and a scoring prediction model as a decoder. They can help the models to capture more heterogeneous interactions, enabling the entity and relation embeddings to learn richer semantic information. Therefore, multi-relation representation learning is helpful for technology opportunity prediction in more accurate rankings.

The results also indicate that CompGCN outperforms RelGCN in the semantic representation of multi-relational data. Specifically, the CompGCN + ConvE model achieves improvements of 1.3% in MRR, 2.1% in hits@1, and 2.8% in hits@3 compared to the RelGCN + ConvE model. This advantage can be attributed to different strategies for handling various relation types in SAO. RelGCN employs separate weight matrices to aggregate information, whereas CompGCN integrates entity and relation representations using combination operations such as circular convolution and addition. This fusion mechanism enables CompGCN to flexibly represent complex semantic relationships between entities and relations, thereby acquiring richer semantic information and achieving superior performance in multi-relational representation learning.

Comparison on different prediction models

To further explore the technology prediction performance of different knowledge graph completion models, this study selects TransE, DistMult and ConvE model for experimental comparison. TransE models relations as translation operations between entities in vector semantic spaces. DistMult utilizes the bilinear scoring function approach for calculating the scores of triples. The result is shown in Table 8, in which the numbers emphasized in italics correspond to the highest value in each column.

The result indicates that ConvE outperforms TransE, DistMult in terms of MRR and hits@n for technology opportunity prediction. Specifically, the ConvE model performs best on the MRR, i.e., 14.0% higher than TransE and 0.5% higher than DistMult. It also performs best on the other metrics, e.g., 8.5% and 19.1% higher than TransE and not much different from DistMult on hits@1 and hits@3 respectively. These improvements may be attributed to that the ConvE can effectively capture heterogeneous feature interactions preserved in entity and relation embeddings by introducing a convolutional layer. While TransE and DistMult are relatively simple, with fewer parameters and capture less complex information compared to the neural network-based model. However, we also found that

Table 7 Performance comparison with or without multi-relation representation learning

Model	MRR	Hits@1	Hits@3	Hits@5	Hits@10	Hits@20
ConvE	0.211	0.129	0.225	0.288	0.387	0.481
RelGCN + ConvE	0.218	0.136	0.243	0.309	0.394	0.490
CompGCN + ConvE	*0.231*	*0.157*	*0.271*	*0.336*	*0.406*	*0.493*

Table 8 Performance comparison on different prediction models

Model	MRR	hits@1	hits@3	hits@5	hits@10	hits@20
CompGCN + TransE	0.091	0.072	0.080	0.083	0.096	0.201
CompGCN + DistMult	0.226	0.156	0.248	0.304	0.395	0.480
CompGCN + ConvE	*0.231*	*0.157*	*0.271*	*0.336*	*0.406*	*0.493*

our results do not perform as well as some knowledge graph completion tasks on general domain datasets, such as the knowledge graph completion experiments on FB15K-237 and WN18RR (Yang et al., 2024; Zhu, 2024). The reason may be that the sample size of our SAO dataset is much smaller than that of the general domain dataset, while the number of actions in our dataset is larger than them. Specifically, our dataset contains 7653 SAOs and 239 actions, while FB15K-237 and WN18RR contain 310,116 SAOs, 237 actions and 93,003 SAOs, 11 actions respectively.

In conclusion, several comparative experiments show that our model composed by a multiple relations graph neural network (CompGCN) as an encoder and a prediction (ConvE) as a decoder, outperforms other methods significantly in predicting technology opportunities.

Case study on predicted technology opportunities

To further illustrate the effectiveness of our method, we analyse some cases which use S and A to predict O. Consistent with the previous experiment, we use the data between 2017 and 2020 to predict the technology opportunities in 2021. By sorting the scores of predicted technical entities (O) in descending order, we select the top 15 ranked technical triplets for detailed analysis. These results are shown in Table 9, in which the underlined technical triples indicate that the predictions are correct.

The ID “1”, “4”, and “7” convey the similar meaning, it indicates that hand-held device systems, computer systems, and kiosks use speech processing. This usage is truly observed in the patents “US2022036893-A1” and “US2021398525-A1” applied in 2021. The abstract of the patent primarily refers to “The speech recognition combined with natural language understanding processing techniques enable speech-based user control of a computing device to perform tasks based on the user’s spoken commands. The speech processing is used by computers, hand-held devices, telephone computer systems, kiosks, and a wide variety of other devices to improve human–computer interactions”.

The ID “6” indicates that user information is converted into voice data, specifically voice diagnosis data, as founds in patents “CN112802598-A” and “CN113555133-A”. These patents primarily involve processes such as “the voice diagnosis and treatment data are transferred to text data by automatic speech recognition engine”, “collecting users diagnostic voice data in a medical diagnosis process”, and “human users transmitted diagnostic voice data, and storing question voice data and structured data in preset database”. However, the accuracy of predicted technology opportunities is insufficient, as they favor the connection of more common technical entities such as “voice data” rather than “voice diagnosis data”.

In addition, the ID “11” is referenced in the patent “US2021233515-A1” in 2021. This patent proposes a method of using language information to adjust acoustic scores, where

Table 9 The result of top 15-ranked emerging technology opportunities

ID	S	A	True "O"	Predicted "O"
1	Computer systems and kiosks	Operate	Speech processing	Speech processing
2	Wide variety devices	Obtain	Human–computer interactions	Human–computer interactions
3	Classification model	Use	Sample data	Recognition data
4	Computer systems and kiosks	Use	Speech processing	Speech processing
5	Recognition model	Obtain	Speech segment	Current set parameter values
6	Human user	Transmit	Voice diagnosis data	Voice data
7	Hand-held devices systems	Obtain	Speech processing	Speech processing
8	Recognition module	Connect	ASR module	Voice command
9	Computers devices	Obtain	Speech processing	Speech processing
10	Computers devices	Identify	Human–computer interactions	Speech processing
11	Language information	Adjust	Acoustic score	Acoustic score
12	Recognition module	Receive	Recognition engine	Voice information
13	Configurable resources	Improve	Minimal management effort	Provider service
14	Word database	Perform	Hot words	Hot word
15	Audio data	Receive	Text string	Audio data

"language information is used to adjust the acoustic score by considering what sounds and words are used in context with each other".

In summary, when using the model to predict technology opportunities in 2021 based on data from 2017 to 2020, it is possible to predict some correct combinations and closely related technical entities. These predicted technology opportunity combinations exhibit relations between technical entities, resulting in more specific technology opportunities. Additionally, the results indicate that the model assigns higher scores to frequently technical entities with stronger associations. Consequently, this tendency may lead to the prediction of previously seen technical combinations rather than the latest technology opportunity combinations.

Conclusion

This paper proposes an approach to predict SAO triplets as technology opportunity from the perspective of SAO extraction and representation. The SAO structures, obtained through dependency parsing analysis, effectively reveal semantic information about technical elements and their relationships. Moreover, CompGCN model integrates relation features into entity features to effectively learn semantic representations of both entities and relationships. Furthermore, ConvE is employed to capture heterogeneous feature interactions within entity and relation embeddings, facilitating the prediction of new technology triplets as technology opportunities. This method enhances the understanding of how multi-relational semantic information can be effectively utilized in technology prediction. By integrating SAO structure extraction with embedding and

interaction modelling techniques, we establish a framework to capture and represent complex relationships among technical elements. This framework provides insights for future advances in semantic representation and predictive analysis.

According to the empirical study, the results have proved the effectiveness of the proposed method. In terms of extracting technical elements and multi-action prediction, SAO structure provides a more detailed description of technology than the keywords extracted from the same patents. Thus, SAOs can reveal how technical elements interact with each other and further illustrate the specific relationships within predicted technical element combinations. Moreover, comparative experiments between multiple actions and a single relation show that incorporating multiple relation information leads to improved prediction performance, achieving a 12.3% increase in MRR. This improvement can be attributed to the fact that replacing all relations with a single type of relation would result in the loss of the richness of the original data, thereby limiting the learning ability of the model. In contrast, adding multiple actions into the model enhances contextual information, enabling the model to effectively encode richer relationships between entities.

In terms of model selection and application, the use of CompGCN led to a 2% improvement in MRR, both in models with and without multi-relational graph neural network representations. This improvement suggests that CompGCN more effectively represents the semantic information embedded in the multiple relationships of the SAO structure. Therefore, multi-relation representation learning is helpful for technology opportunity prediction. Moreover, among the different scoring functions, ConvE outperforms others with an MRR of 0.231 and a hit@5 of 0.336. This superiority is attributed to its effective capture of heterogeneous feature interactions preserved in entity and relation embeddings by introducing a convolutional layer. Overall, this method, which utilizes CompGCN as the representation module and ConvE as the prediction module, outperforms existing technology opportunity prediction methods, providing more accurate and comprehensive results. The proposed approach can be extended to other domains beyond speech recognition. The findings of this study may help researchers, investors, and business managers in other fields identify emerging or prominent technology opportunities. Additionally, this approach provides R&D departments and researchers with potential research directions, facilitates the identification of innovation opportunities in competitive environments, and offers strategic insights for decision-makers.

This method provides a reference for technology opportunity prediction. However, this is only a preliminary study, and further improvements are needed in the following directions. First, this study only uses patents as the data source, without considering the role of scientific and market-related data for technology opportunity identification. In the future, it may be beneficial to incorporate multiple data sources, such as academic papers, patent transactions, and market trade information, to construct more comprehensive indices for technology opportunity identification. Second, this study solely focuses on exploring technology opportunities within a specific domain. This makes our dataset have fewer entities but more actions compared to general datasets. In the future, it would be worthwhile to explore innovative technology opportunities in other domains or cross-domains and increase data size. Finally, although this study has selected representative algorithms to perform semantic representation and technology opportunity prediction, but other recent methods are still emerging that provide better performance of semantic representation and knowledge graph completion. Therefore, it is necessary to explore more recent methods of SAO embedding and prediction for comparison in the future.

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Declarations

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